Exercise C - II

1. 3 students {A, B, C} tackle a puzzle together and offers a solution upon which majority of the 3 agrees. Probability of A solving the puzzle correctly is p. Probability of B solving the puzzle correctly is also p. C is a dumb student who randomly supports the solution of either A or B. There is one more student D, whose probability of solving the puzzle correctly is once again, p. Out of the 3 member team {A, B, C} and one member team {D}, Which one is more likely to solve the puzzle correctly.

3 विद्यार्थी {A, B, C} किसी पहैली के एक साथ हल करते है तथा पहैली पूर्ण कहलाती है, यदि हल पर तीन विद्यार्थीयों बहुमत का उत्तर सही है। A तथा B द्वारा पहैली को हल करने कि प्रायिकता p है तथा C कमजोर विद्यार्थी है तथा वह A अथवा B के हल पर यादृच्छया समर्थन करता है। एक और विद्यार्थी D है जिसके द्वारा पहेली को हल करने की प्रायिकता भी p है। तीन सदस्यों की टीम {A, B, C}तथा एक विद्यार्थी की टीम {D}में से पहेली को हल करने की प्रायिकता किसकी ज्यादा होगी।

Ans. Both are equally likely

Sol. Let the cases
Case-I : A solves puzzle correctly and B solves puzzle correctly and C supports either of A or B, then probability $= p \times p \times 1$
Case-II : A solve puzzle correctly and B solves puzzle incorrectly and C supports A

then probability $= p \times (1-p) \times \frac{1}{2} = \frac{1}{2}p - \frac{1}{2}p^2$

Case-III : A solve puzzle incorrectly and B solves puzzle correctly and C supports B

then probability $= (1-p) \times p \times \frac{1}{2} = \frac{1}{2}p - \frac{1}{2}p^2$

Then the probability of out of 3 students {A, B, C} solve the puzzle $= p^2 + \frac{1}{2}p - \frac{1}{2}p^2 + \frac{1}{2}p - \frac{1}{2}p^2$
$= p$

Probability of solving the puzzle by D is $= p$.
So that bnoth are equally likely.

2. A uniform unbised die is constructed in the shape of a regular tetrahedron with faces numbered 2, 2, 3 and 4 and the score is taken from the face on which the die lands. If two such dice are thrown together, find the probability of scoring.

(i) Exactly 6 on each of 3 successive throws.

(ii) More than 4 on at least one of the three successive throws

एक समचतुष्कफलक के आकार का पासा बनाया जाता है। जिसके फलकों पर 2, 2, 3 तथा 4 अंकित है तथा जिस फलक से पासा गिरता है उसका अंक लिया जाता है। यदि इस तरह के दो पासे लिये जाते है। तो प्रायिकता ज्ञात कीजिये।

(i) जमीन पर दोनों पासों को लगातार तीन बार फैंकने पर प्रत्येक बार अंको का योग 6 हो

(ii) दोनों पासों को लगातार तीन बार फैंकने पर अंको को योग कम से कम एक बार 4 से अधिक हो

Ans. (i) $\frac{125}{16^3}$; (ii) $\frac{63}{64}$

Sol. n(5) = total no. of outcomes = $4 \times 4 = 16$
n (sum of numbers = 6) = 5 {(2, 4) (4, 2) (2′, 4) (2, 4′) (3, 3)}

P (exactly 6 one each successive 3 throws) = $\frac{5}{16} \times \frac{5}{16} \times \frac{5}{16} = \frac{125}{4096}$

n (sum of number = 4) = {(2, 2) (2′,2) (2,2′) (2′,2′)}
P (more than 4 on at least one of the three successive throws) = 1 – P (4 on 3 throws)

$= 1 - \frac{4}{16} \times \frac{4}{16} \times \frac{4}{16} = \frac{63}{64}$

3. Two cards are drawn from a well shuffled pack of 52 cards. Find the probability that one of them is a red card and the other is a queen.

52 पत्तों की ताश कि गड्डी में दो पत्ते चुने जाते है तो उनमें से एक पत्ता लाल तथा दूसरा रानी होने की प्रायिकता ज्ञात कीजिये।

Ans. 101/1326

Sol. $E_1 \rightarrow$ Both red queen

$E_2 \rightarrow$ 1 red queen other normal red

$E_3 \rightarrow$ 1 black queen any red

$$\text{Prob} = \frac{{}^2C_2 + {}^2C_1 \times {}^{24}C_1 + {}^2C_1 \times {}^{26}C_1}{{}^{52}C_2} = \frac{1 + 48 + 52}{26 \times 51} = \frac{101}{1326}$$

4. 16 players take part in a tennis tournament. The order of the matches is chosen at random. There is always a player better than another one, the better wins. Find :

(a) The probability that all the 4 best players reach the semifinals.

(b) The probability that the sixth best reaches the semifinals.

एक टेनिस प्रतियोगिता में 16 खिलाड़ी भाग लेते है। खिलाड़ी एक दूसरे से अच्छा खेलता है, तथा अच्छे वाले खिलाड़ी ही जीतता है :

(a) तो प्रायिकता ज्ञात कीजिये जब 4 सबसे बढ़िया खिलाड़ी सेमीफाइनल में पहुँचते है

(b) तो प्रायिकता ज्ञात कीजिये जब छठा सबसे बढ़िया खिलाड़ी सेमीफाइनल में पहुँचता है

Ans. (a) $\frac{64}{455}$; (b) $\frac{24}{91}$

Sol. Total no. of ways of division of 16 players into 4 groups each containing 4 players

$$= \frac{\lfloor 16}{(\lfloor 4)^4 . \lfloor 4} = n(5)$$

No. of ways in which 16 players can be divided into 4 groups so that 4 best players go in 4 different groups

$$= \frac{\lfloor 12}{(\lfloor 3)^4 . \lfloor 4} \times \lfloor 4$$

(i) P (all the best players reach the semifinal) $= \dfrac{\dfrac{\lfloor 12}{(\lfloor 3)^4}}{\dfrac{\lfloor 16}{(\lfloor 4)^5}} = \dfrac{64}{455}$

(ii) sixth best player reaches to the semifinal exactly when other 3 players in his group of four are not in top 6, i.e. are in the set {7, 8, 9, ... 16}.
There are total $^{15}C_3$ ways to select the othe three players in the group, of these $^{10}C_3$ consist entirely of players not in the top 6.

So that probability that sixth best player each in semifinal is $= \dfrac{^{10}C_3}{^{15}C_3} = \dfrac{24}{91}$

5. The probabilities that three men hit a target are, respectively, 0.3, 0.5 and 0.4. Each fires once at the target. (As usual, assume that the three events that each hits the target are independent)

(a) Find the probability that they all : **(i)** hit the target; **(ii)** miss the target

(b) Find the probability that the target is hit : **(i)** at least once, **(ii)** exactly once

(c) If only one hits the target, what is the probability that it was the first man?

तीन व्यक्तियों द्वारा लक्ष्य को साधने की प्रायिकता 0.3, 0.5 और 0.4 है। प्रत्येक व्यक्ति एक निशाना साधता है। (माना तीनों घटनाऐं जिसमें प्रत्येक व्यक्ति निशाना साधता है, स्वतन्त्र है)

(a) प्रायिकता ज्ञात कीजिए जबकि सभी : **(i)** निशाना साधते है; **(ii)** निशाना नहीं साधते है।

(b) प्रायिकता ज्ञात कीजिए जबकि : **(i)** कम से कम एक व्यक्ति निशाना साधता है, **(ii)** ठीक एक व्यक्ति निशाना साधता है।

(c) यदि केवल एक निशाना साधता है तो प्रायिकता ज्ञात कीजिए जब वह पहला व्यक्ति है?

Ans. (a) 6%, 21% ; (b) 79%, 44%, (c) 9/44 » 20.45%

Sol. Let the events
A = First men hit the target
P(A) = 0.3
P(A′) = 0.7
B = second men hit the target
P(B) = 0.5

$P(B') = 0.5$
C = third men hit the target
$P(C) = 0.4$
$P(C') = 0.6$
(i) P (All 3 hit the target) = $P(A \cap B \cap C)$
$= P(A)\, P(B)\, P(C)$
$= 0.3 \times 0.5 \times 0.4 = 0.06$
P (All miss the target) = $P(A' \cap B' \cap C')$
$= P(A')\, P(B')\, P(C')$
$= 0.7 \times 0.5 \times 0.6 = 0.21$
(ii) P (atleast one hit the target) = $P(A \cup B \cup C)$
$= 1 - P(A' \cap B' \cap C)$
$= 1 - 0.21 = 0.79$
P (exactly one hit the target) = $P(A \cap B' \cap C') + P(A' \cap B \cap C') + P(A' \cap B' \cap C)$
$= P(A)\, P(B')\, P(C') + P(A')\, P(B)\, P(C') + P(A')\, P(B')\, P(C)$
$= 0.3 \times 0.5 \times 0.6 + 0.7 \times 0.5 \times 0.6 + 0.7 \times 0.5 \times 0.4$
$= 0.44$
(iii) If only one hits the target, then the probability that it was the first man is

$$= \frac{P(A \cap B' \cap C')}{P(A \cap B' \cap C') + P(A' \cap B \cap C') + P(A' \cap B' \cap C)}$$

$$= \frac{0.3 \times 0.5 \times 0.6}{0.44} = \frac{9}{44}$$

6. A pair of students is selected at random from a probability class. The probability that the pair selected will consist of one male and one female student is $\frac{10}{19}$. Find the maximum number of students the class can contain.

एक प्रायिकता कि कक्षा में से एक विद्यार्थी का जोड़ा यादृच्छया चुना जाता है, तो चुने गये जोड़े में एक पुरूष तथा एक महिला होने की प्रायिकता $\frac{10}{19}$ है, तो कक्षा में विद्यार्थी की अधिकतम संख्या ज्ञात कीजिये।

Ans. 20

Sol. Let number of male student = x
number of female student = y

P (Pair consist one male and one female) $\Rightarrow P(MF) + P(FM) = \frac{10}{19}$

$$\Rightarrow \frac{x}{x+y} \times \frac{y}{x+y-1} + \frac{y}{x+y} \times \frac{x}{x+y-1} = \frac{10}{19}$$

$$\Rightarrow \frac{2xy}{(x+y)(x+y-1)} = \frac{10}{19}$$

Let $x + y = m$

$$\frac{2x(m-x)}{m(m-1)} = \frac{10}{19}$$

$19mx - 19x^2 = 5m^2 - 5m$

$$\Rightarrow x^2 - mx + \frac{5}{19}(m^2 - m) = 0$$

$D \geq 0$

$b^2 - 4ac \geq 0$

$$m^2 - 4 \times \frac{5}{19}(m^2 - m) \geq 0$$

$19m^2 - 20m^2 + 20m \geq 0$

$m^2 - 20m \leq 0$

$m(m - 20) \leq 0$

So that maximum number of student = 20

7. N fair coins are flipped once. The probability that at most 2 of the coins show up as heads is $\frac{1}{2}$. Find the value of N.

छ सिक्के एक बार उछाले जाते है। अधिकतम दो सिक्को पर चिट आने कि प्रायिकता $\frac{1}{2}$ हो तो N का मान ज्ञात कीजिये।

Ans. 5

Sol. P (atmost two coins show up heads) = $\frac{1}{2}$

$$P(X = 0) + P(X = 1) + P(X = 2) = \frac{1}{2}$$

$$^nC_0\left(\frac{1}{2}\right)^n + {}^nC_1\left(\frac{1}{2}\right)^n + {}^nC_2\left(\frac{1}{2}\right)^n = \frac{1}{2}$$

$$1 + n + n\cdot\frac{(n-1)}{2} = 2^{n-1}$$

$2 + 2n + n^2 - n = 2^n$

$n^2 + n + 2 = 2^n$

So taht n = 5

8. Three shots are fired independently at a target in succession. The probabilities that the target is hit in the first shot is 1/2 , in the second 2/3 and in the third shot is 3/4. In case of exactly one hit , the probability of destroying the target is 1/3 and in the case of exactly two hits, 7/11 and in the case of three hits is 1.0. Find the probability of destroying the target in three shots.

स्वतन्त्र रूप से एक लक्ष्य पर तीन लगातार निशाने साधे जाते है। पहले निशाने में लक्ष्य को साधने की प्रायिकता 1/2 है तथा दूसरे में 2/3 तथा तीसरे में 3/4 है। यदि केवल एक लक्ष्य साधा जाता है तो लक्ष्य के नष्ट होने की प्रायिकता 1/3 तथा यदि ठीक दो लक्ष्य साधते है तो प्रायिकता 7/11 तथा यदि तीनों लक्ष्य साधते हैं तो प्रायिकता 1 है। तो लक्ष्य को तीन निशानों में नष्ट होने की प्रायिकता ज्ञात कीजिए।

Ans. $\frac{5}{8}$

Sol. A = target hit in 1st shot
B = target hit in 2nd shot
C = target hit in 3rd shot
E_1 = Target destroyed in exactly one shot
E_2 = Target destroyed in exactly two shot
E_3 = Target hit in exactly three shot

$$P(E_1) = P\left(E_1 A\overline{BC} \cup E_1 \overline{AB}C \cup E_1 \overline{A}B\overline{C}\right)$$

$$= \frac{1}{3}\left[\frac{1}{2}\cdot\frac{1}{3}\cdot\frac{1}{4} + \frac{1}{2}\cdot\frac{1}{3}\cdot\frac{3}{4} + \frac{1}{2}\cdot\frac{2}{3}\cdot\frac{1}{4}\right] = \frac{1}{12}$$

$$P(E_2) = P\left(E_2 \overline{A}BC + E_2 A\overline{B}C + E_3 AB\overline{C}\right)$$

$$= \frac{7}{11}\left(\frac{1}{2}\cdot\frac{2}{3}\cdot\frac{3}{4} + \frac{1}{2}\cdot\frac{2}{3}\cdot\frac{1}{4} + \frac{1}{2}\cdot\frac{1}{3}\cdot\frac{3}{4}\right) = \frac{7}{24}$$

$$P(E_3) = P(E_3 ABC) = 1\cdot\frac{1}{2}\cdot\frac{2}{3}\cdot\frac{3}{4} = \frac{1}{4}$$

$$P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) + P(E_3)$$

$$= \frac{1}{12} + \frac{7}{24} + \frac{1}{4} = \frac{5}{8}$$

9. In a game of chance each player throws two unbiased dice and scores the difference between the larger and smaller number which arise. Two players compete and one or the other wins if and only if he scores atleast 4 more than his opponent. Find the probability that neither player wins.

एक खेल में प्रत्येक खिलाड़ी दो पासें फेंकता है तथा उसका स्कोर बड़ी संख्या तथा छोटी संख्या में अन्तर है। दो खिलाड़ी खेलते है। यदि केवल एक व्यक्ति जीतता है, यदि वह अपने प्रतिद्वंदी से कम से कम चार अंक ज्यादा लाता है। तो कोई भी खिलाड़ी के नहीं जीतने की प्रायिकता ज्ञात कीजिए।

Ans. 74/81

Sol. The possible scores are {0, 1, 2, 3, 4, 5}
Let P(X) be the probability of scores

$$P(5) = \{(6,1)\ (1,6)\} = \frac{2}{36}$$

$$P(4) = \{(2,6)\ (6,2)\ (1,5)\ (5,1)\} = \frac{4}{36}$$

$P(1) = \{(1,2)\ (2,1)\ (3,2)\ (3,4)\ (4,3)\ (4,5)\ (5,4)\ (5,6)\ (6,5)\} = \frac{10}{36}$

$P(0) = \{(1,1)\ (2,2)\ (3,3)\ (4,4)\ (5,5)\ (6,6)\} = \frac{6}{36}$

Probability of win $P(\geq 4) = [P(5)\,P(1) + P(4)\,P(0) + P(5)\,P(0)\ ] \times 2! \rightarrow$ Arrangement of scores in P_1 & P_2

$$= 2 \times \left(\frac{2}{36} \times \frac{10}{36} + \frac{4}{36} \times \frac{6}{36} + \frac{2}{36} \times \frac{6}{36} \right) = \frac{7}{81}$$

$$P\ (\text{neither wins}) = 1 - \frac{7}{81} = \frac{74}{81}$$

10. There are two packs A and B of 52 playing cards. All the four aces from the pack A are removed whereas from the pack B, one ace, one king, one queen and one jack is removed. One of these two packs is slected randomly and two cards are drawn simultaneously from it, and found to be a pair (i.e. both have same rank e.g. two 9's or two king etc). Find the probability that the pack A was selected.

52 पत्तों की दो ताश की गड्डी A तथा B है। ताश A में चारों इक्के तथा ताश B में से 1 राजा, 1 रानी, 1 गुलाम तथा एक इक्का हटाया जाता है। यदि इन ताशों में से कोई भी एक ताश चुनी जाती है। तो इसमें से एक साथ दो पत्ते निकाले जाते हैं तथा युग्म पाया जाता है। (दोनों समान रेंक के है। उदाहरण के लिए दो 9's, दो राजा etc) तो प्रायिकता ज्ञात कीजिए कि ताश A चुनी जाती है।

Ans. $\frac{12}{23}$

Sol. Let the events

A = Pack of cards from 4 aces are removed

B = Pack of cards in which one ace, one king, one queen, one jack is removed

$P(A) = P(B) = \frac{1}{2}$

C = event of one pair is selected

If both drawn card found to be a pair, then the probability that pack A was selected

$$P\left(\frac{A}{C}\right) = \frac{P(A)P\left(\frac{C}{A}\right)}{P(A)P\left(\frac{C}{A}\right) + P(B)P\left(\frac{C}{B}\right)}$$

$$= \frac{\frac{1}{2} \times {}^{12}C_1\, {}^{4}C_2}{\frac{1}{2} \times {}^{12}C_1\, {}^{4}C_2 + \frac{1}{2}\left({}^{4}C_1\, {}^{3}C_2 + {}^{9}C_1\, {}^{4}C_2\right)}$$

$$= \frac{72}{72+12+54} = \frac{72}{138}$$

$$= \frac{12}{23}$$

11. During a power blackout, 100 persons are arrested on suspect of looting. Each is given a polygraph test. From past experience it is known that the polygraph is 90% reliable when administered to a guilty person and 98% reliable when given to some one who is innocent. Suppose that of the 100 persons taken into custody, only 12 were actually involved in any wrong doing. If the probability that a given suspect is innocent given that the polygraph says he is guilty is $\frac{a}{b}$ where *a* and *b* are relatively prime, find the value of (a + b).

पूरे शहर में बिजली जाने के दौरान 100 संदिग्ध व्यक्ति गिरफ्तार किये जाते है तथा प्रत्येक पर पालिग्राफ टेस्ट किया जाता है। पुराने प्रयोगों से पता चलता है कि पालिग्राफ टेस्ट 90% सही है, जबकि व्यक्ति दोषी है तथा 98% सही है जबकि व्यक्ति दोषी नहीं है। माना 100 व्यक्ति जेल में ले जाये जाते है, केवल 12 व्यक्ति ही गलत कामों में लिप्त है तो प्रायिकता ज्ञात कीजिए कि संदिग्ध दोषी नहीं है, यदि दिया गया है कि व्यक्ति का पोलीग्राफ दोषी बताता है। उसकी प्रायिकता $\frac{a}{b}$ है जहां *a* तथा *b* सापेक्ष अभाज्य है तो (a + b) का मान है।

Ans. 179

Sol. Let the events

A = person actually involved in wrong doing

$$P(A) = \frac{12}{100}$$

P(B) = person is innoccent

$$P(B) = \frac{88}{100}$$

T = Reliability of polygraph test

$$P\left(\frac{T}{A}\right) = \frac{90}{100}$$

$$P\left(\frac{T}{B}\right) = \frac{2}{100}$$

If polygraph test found a person is guilty, then the probability that person is innocent

$$P\left(\frac{A}{T}\right) = \frac{P(B)P\left(\frac{T}{B}\right)}{P(A)P\left(\frac{T}{A}\right)+P(B)P\left(\frac{T}{B}\right)}$$

$$= \frac{\frac{88}{100}\times\frac{2}{100}}{\frac{12}{100}\times\frac{90}{100}+\frac{88}{100}\times\frac{2}{100}}$$

$$= \frac{22}{157}$$

So that a + b = 22 + 157 = 179

12. A biased coin which comes up heads three times as often as tails is tossed. If it shows heads, a chip is drawn from urn–I which contains 2 white chips and 5 red chips. If the coin comes up tails, a chip is drawn from urn–II which contains 7 white and 4 red chips. Given that a red chip was drawn, what is the probability that the coin came up heads?

एक सिक्के को उछालने पर चिट आने कि प्रायिकता पट से 3 गुना है। यदि यह चिट दिखाता है तो बक्से –I में से एक चिप निकाली जाती है जिसमें दो सफेद तथा 5 लाल चिप है। यदि पट आता है तो बक्से–II में से एक चिप निकाली जाती है जिसमें 7 सफेद तथा 4 लाल चिप gSA fn; kx; kgS fd fudkyh xbZ chp ykyh gS] ;fn ;g fpV vkrk gS rks ;g izkf;drk Kkr dhft ;s fd fl Dd ls dksu lh Nkyh usi j fpV vk;k gS

Ans. 165/193

Sol. Given that P(Heads) = 3 P(Tails)

P(H) = 3 P(T)

$\therefore$ P(H) + P(T) = 1

$$P(T) = \frac{1}{4}$$

$$P(H) = \frac{3}{4}$$

R = Event of red chip is drawn

$$P\left(\frac{R}{H}\right) = \frac{5}{7}$$

$$P\left(\frac{R}{T}\right) = \frac{4}{11}$$

If a red chip is drawn, then the probability that coin comes up with head

$$P\left(\frac{H}{R}\right) = \frac{P(H)P\left(\frac{R}{H}\right)}{P(H)P\left(\frac{R}{H}\right)+P(T)P\left(\frac{R}{T}\right)}$$

$$= \frac{\frac{3}{4}\times\frac{5}{7}}{\frac{3}{4}\times\frac{5}{7}+\frac{1}{4}\times\frac{4}{11}}$$

$= \frac{165}{193}$

13. A box contains three coins two of them are fair and one two – headed. A coin is selected at random and tossed. If the head appears the coin is tossed again, if a tail appears, then another coin is selected from the remaining coins and tossed.

(i) Find the probability that head appears twice.

(ii) If the same coin is tossed twice, find the probability that it is two headed coin.

(iii) Find the probability that tail appears twice.

एक बक्से में तीन सिक्के हैं जिनमें दो साधारण तथा एक सिक्के में दो पट्ट हैं। कोई भी एक सिक्का चुना जाता है तथा उसे उछाला जाता है। यदि सिक्के पर पट्ट आता है तो उसे फिर से उछाला जाता है। यदि चित्त आता है तो बाकी बचे हुए सिक्के में से कोई भी एक सिक्का चुना जाता है और उछाला जाता है तो

(i) प्रायिकता ज्ञात कीजिए जबकि दो बार पट्ट आता है।

(ii) यदि एक ही सिक्के को उछाला जाता है तो प्रायिकता ज्ञात कीजिए जबकि यह दो पट्ट वाला सिक्का है।

(iii) चित्त के दो बार आने की प्रायिकता ज्ञात कीजिए।

Ans. (i)1/2 (ii) 2/3, (iii)1/12

Sol. A box contains three coins, two of them fair and one two-headed. A coin is selected at random and tossed. If heads appear the coin is tossed again; if tails appear then another coin is selected from the two remaining coins and ossed.

(i) find the probability that heads appear twice.

(ii) if the same coni is tossed twice, find the probability that it is the two-headed coin.

(iii) find the probability that tails appear twice.

In each case we select a coin, toss a coin, select a coin, then toss a coin. Sometimes it is certain (probability = 1) which coin we select, sometimes not. Somtimes it's certain that we 11 get a head (probability = 1), some times not.

There are six cases. I can't make a tree diagram here, but you can and should. Here are the six cases :

	Select(Prob.)	**Toss(Prob.)**	**Select(Prob.)**	**Toss(Prob.)**	**Probability of possibility**
1.	$\text{Fair}\left(\frac{2}{3}\right)$	$\text{Head}\left(\frac{1}{2}\right)$	Fair(1)	$\text{Head}\left(\frac{1}{2}\right)$	$\left(\frac{2}{3}\right)\left(\frac{1}{2}\right)(1)\left(\frac{1}{2}\right)=\frac{1}{6}$
2.	$\text{Fair}\left(\frac{2}{3}\right)$	$\text{Head}\left(\frac{1}{2}\right)$	Fair(1)	$\text{Tail}\left(\frac{1}{2}\right)$	$\left(\frac{2}{3}\right)\left(\frac{1}{2}\right)(1)\left(\frac{1}{2}\right)=\frac{1}{6}$
3.	$\text{Fair}\left(\frac{2}{3}\right)$	$\text{Tail}\left(\frac{1}{2}\right)$	$\text{Fair}\left(\frac{1}{2}\right)$	$\text{Head}\left(\frac{1}{2}\right)$	$\left(\frac{2}{3}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)=\frac{1}{12}$

4. Fair $\left(\frac{2}{3}\right)$ Tail $\left(\frac{1}{2}\right)$ Fair $\left(\frac{1}{2}\right)$ Tail $\left(\frac{1}{2}\right)$ $\left(\frac{2}{3}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)=\frac{1}{12}$

5. Fair $\left(\frac{2}{3}\right)$ Tail $\left(\frac{1}{2}\right)$ 2 headed $\left(\frac{1}{2}\right)$ Head (1) $\left(\frac{2}{3}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)(1)=\frac{1}{6}$

6. 2 headed $\left(\frac{1}{3}\right)$ Head(1) 2 headed (1) Head (1) $\left(\frac{1}{3}\right)(1)(1)(1)=\frac{1}{3}$

Notice that the sum of probabilities of all six cases is 1, i.e.,

$$\frac{1}{6}+\frac{1}{6}+\frac{1}{12}+\frac{1}{12}+\frac{1}{6}+\frac{1}{3}=\frac{2}{12}+\frac{2}{12}+\frac{1}{12}+\frac{1}{12}+\frac{2}{12}+\frac{4}{12}=\frac{12}{12}=1$$

(i) find the probability that heads appear twice.

P(case 1 OR case 6) = $\frac{1}{6}+\frac{1}{3}=\frac{1}{6}+\frac{2}{6}=\frac{3}{6}=\frac{1}{2}$

(ii) If the same coin is tossed twice, find the probability that it is the two - headed coin.

P(2-headed|tossed twice) = P(case 6|case 1 or case 6) =

$$\frac{P(\text{case 6 AND case 1 or 6})}{P(\text{case 1 OR case 6})}=\frac{P(\text{case 6})}{\frac{1}{2}[\text{from (i)}]}=\frac{\frac{1}{3}}{\frac{1}{2}}=\left(\frac{1}{3}\right)\cdot\left(\frac{2}{1}\right)=\frac{2}{3}$$

(iii) find the probability that tails appear twice.

P(case 4) = $\frac{1}{12}$

Edwin

14. The ratio of the number of trucks along a highway, on which a petrol pump is located, to the number of cars running along the same highway is 3 : 2. It is known that an average of one truck in thirty trucks and two cars in fifty cars stop at the petrol pump to be filled up with the fuel. If a vehicle stops at the petrol pump to be filled up with the fuel, find the probability that it is a car.

एक राजमार्ग पर ट्रकों तथा कारों की संख्याओं का अनुपात 3 : 2 है जिस पर पेट्रोल पम्प है तथा तीस ट्रकों में एक तथा 50 कारों में से दो कारें तेल भरवाने के लिए पेट्रोल पम्प पा सकती हैं। यदि एक वाहन पेट्रोल पम्प पर रूकता है। तो उसके कार होने की प्रायिकता ज्ञात कीजिए।

Ans. $\frac{4}{9}$

Sol. Let the events
A = event of trucks on highway
B = event of cars on highway

$$P(A)=\frac{3}{5}$$

$$P(B)=\frac{2}{5}$$

T = Event of vehicle stops at petrol pump to be filled up with the fuel

$$P\left(\frac{T}{A}\right)=\frac{1}{30}$$

$$P\left(\frac{T}{B}\right)=\frac{2}{50}=\frac{1}{25}$$

If a vehicle stops at petrol pump to be filled up with the fuel, then the probability that it is a car

$$P\left(\frac{B}{T}\right)=\frac{P(B)P\left(\frac{T}{B}\right)}{P(A)P\left(\frac{T}{A}\right)+P(B)P\left(\frac{T}{B}\right)}$$

$$=\frac{\frac{2}{5}\times\frac{1}{25}}{\frac{3}{5}\times\frac{1}{30}+\frac{2}{5}\times\frac{1}{25}}$$

$$=\frac{4}{9}$$

15. Integers a, b, c and d not necessarily distinct, are chosen independently and at random from the set S = {0, 1, 2, 3, ... 2006, 2007}. If the probability that | ad – bc | is even, is $\frac{p}{q}$ where p and q are relatively prime the find the value of (p + q).

पूर्णांक a, b, c और d (अलग अलग होना जरूरी नहीं है) एक समुच्चय S = {0, 1, 2, 3, ... 2006, 2007} में से चुने जाते हैं। यदि | ad – bc | के सम होने की प्रायिकता $\frac{p}{q}$ है जहां p और q परस्पर अभाज्य संख्याऐं है। तो (p + q) का मान ज्ञात कीजिये।

Ans. 13

Sol. S = {0,1,2,3 ..... 2006, 2007}
half of integers are odd and half are even is set S

$P(\text{odd}) = \frac{1}{2}$

$P(\text{even}) = \frac{1}{2}$

$|ad - bc|$ = is odd if

(a) case-I = odd – even

(b) Case-II = even – odd

(a) Case-I = P(odd – even)

= P(odd × odd – even × even) + P(odd × odd – even × odd) + P(odd × odd – odd × eeven)

$$= \frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2} + \frac{1}{2}\times\frac{1}{2} + \frac{1}{2}\times\frac{1}{2} + \frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}$$

$$= \frac{3}{16}$$

(b) Case-II = P(even – odd)

= P(even × even – odd × odd) + P(even × odd – odd × odd) + P(odd × even – odd × odd)

$$= \frac{1}{16} + \frac{1}{16} + \frac{1}{16} = \frac{3}{16}$$

$$P(|ad - bc| \text{ is odd}) = \frac{3}{16} + \frac{3}{16} = \frac{6}{16} = \frac{3}{8}$$

So that

$$P(|ad - bc| \text{ is even}) = 1 - \frac{3}{8} = \frac{5}{8} = \frac{p}{q}$$

Hence p + q = 5 + 8 = 13

16. Eight players $P_1, P_2, P_3, \ldots\ldots\ldots P_8$ play a knock-out tournament. It is known that whenever the players P_i and P_j play, the player P_i will win if $i < j$. Assuming that the players are paired at random in each round, what is the probability that the player P_4 reaches the final.

आठ खिलाड़ी $P_1, P_2, P_3, \ldots\ldots\ldots P_8$ एक नॉक आउट टूर्नामेंट खेलते हैं। यह ज्ञात है कि जब भी व्यक्ति P_i व P_j खेलते हैं तो P_i जीतता है यदि $i < j$. तो खिलाड़ी P_4 के फाइनल में पहुंचने की प्रायिकता ज्ञात कीजिए।

Ans. 4/35

Sol. P_1 is best player.

P_4 win the match if it plays with P_5, P_6, P_7, P_8.

Number of group for which P_1, P_2, P_3 (any one of P_5, P_6, P_7, P_8) in a group and P_4 (any three from P_5, P_6, P_7, P_8) are in one group = ${}^4C_1 = 4$

Total number of two groups each group having 4 players are = $\frac{|\underline{8}}{|\underline{4}|\underline{4}|\underline{2}} = 35$

Required probability = $\frac{4}{35}$

17. A hunter's chance of shooting an animal at a distance 'r' is $\frac{a^2}{r^2}$ $(r > a)$. He fires when $r = 2a$ and if he misses he reloads & fires when $r = 3a, 4a, ...$ If he misses at a distance 'na', the animal escapes. Find the odds against the hunter.

एक शिकारी एक जानवर को 'r' दूरी से निशाना बनाने की प्रायकिता $\frac{a^2}{r^2}$ है जहां $(r > a)$. जब $r = 2a$ है तो वह निशाना साधता है। यदि उसका निशाना चूकता है तो वह पुनः निशाना साधता है। जब $r = 3a, 4a, ...$ यदि वह 'na' दूरी से निशाना चूकता है तथा जानवर भाग जाता है। तो शिकारी के विपक्ष में संयोगानुपात ज्ञात कीजिए।

Ans. $n + 1 : n - 1$

Sol. E_r = hunter shoots the animal at r distance

$P(E) = P(\text{escapes}) = P\left(\overline{E_{2a}}\,\overline{E_{3a}}\,\overline{E_{4a}}.....\overline{E_{na}}\right)$

$$=\left(1-\frac{1}{2^2}\right)\left(1-\frac{1}{3^2}\right).....\left(1-\frac{1}{n^2}\right)$$

$$=\left(\frac{1\cdot3}{2\cdot2}\right)\left(\frac{2\cdot4}{3\cdot3}\right).....\left(\frac{n+1}{n}\cdot\frac{n-1}{n}\right)$$

$$=\frac{n+1}{2n}$$

odd against the hunter $=\frac{n+1}{n-1}$

18. In a tournament, team X, plays with each of the 6 other teams once. For each match the probabilities of a win, draw and loss are equal. Find the probability that the team X, finishes with more wins than losses.

एक प्रतियोगिता में टीम X अन्य 6 टीमों से प्रत्येक केवल एक मैच खेलती है। मैच के जीतने, हारने व ड्रा होने की प्रायिकता समान है। टीमें X के हारने से ज्यादा मैच जीतने की प्रायिकता ज्ञात कीजिए।

Ans. $\frac{98}{243}$

Sol. Total number of result $= 3^6 = 729$

n(E) = Number of ways in which team X finishes with more wins than losses $= \frac{1}{2}$ (729 – Number of ways in which team X finishe with equal number of wins and losses)

Case-I : 0 Draw, 3 wins, 3 losses (WWWLLL)

$$=\frac{\lfloor 6}{\lfloor 3 \lfloor 3}=20$$

Case-II : 1 win, 1 loss, 4 draws (WLDDDD)

$$= \frac{\lfloor 6}{\lfloor 1 \lfloor 1 \lfloor 4} = 30$$

Case-III : 2 wins, 2 losses, 2 draws (WWLLDD)

$$= \frac{\lfloor 6}{\lfloor 2 \lfloor 2 \lfloor 2} = 90$$

Case-IV : No win and no loss, 6 draws = (DDDDDD)

$$= \frac{\lfloor 6}{\lfloor 6} = 1$$

So that $n(E) = \frac{1}{2}\left(729 - \left(20 + 30 + 90 + 1\right)\right)$

$= 294$

Required probability $= \frac{294}{729} = \frac{98}{243}$

19. With respect to a particular question on a multiple choice test (having 4 alternatives with only 1 correct) a student knows the answer and therefore can eliminate 3 of the 4 choices from consideration with probability 2/3, can eliminate 2 of the 4 choices from consideration with probability 1/6, can eliminate 1 choice from consideration with probability 1/9, and can eliminate none with probability 1/18. If the student knows the answer, he answers correctly, otherwise he guesses from among the choices not eliminated. If the answer given by the student was found correct, then the probability that he knew the answer is $\frac{a}{b}$ where a and b are relatively prime. Find the value of (a + b).

किसी बहुविकल्पीय प्रश्न के चार विकल्प है, जिनमें से केवल एक सही है। एक विद्यार्थी उत्तर जानता है तथा उसके 4 विकल्पों में से तीन विकल्प हटाने की प्रायिकता 2/3 है तथा 4 में से 2 विकल्प हटाने की प्रायिकता 1/6 है तथा एक विकल्प हटाने की प्रायिकता 1/9 है तथा एक भी विकल्प नहीं हटाने की प्रायिकता 1/18 . यदि विद्यार्थी उत्तर जानता है। तो उसका सही उत्तर देता है अन्यथा वह एक तुक्का मारता है जो विकल्प उसने हटाया नहीं है। यदि विद्यार्थी द्वारा दिया गया उत्तर सही है। तो उसके उत्तर ज्ञात होने की प्रायिकता $\frac{a}{b}$ है जहां a तथा b सापेक्ष अभाज्य संख्याऐं हैं तो (a + b) का मान ज्ञात कीजिए।

Ans. 317

Sol. A = student eliminate 3 of the 4 choices

$$P = \frac{2}{3}$$

B = student eliminate 2 of the 4 choices

$$P(B) = \frac{1}{6}$$

C = student eliminate 1 of the 4 choices

$$P(C)=\frac{1}{9}$$

D = student do not eliminate any choice

$$P(D)=\frac{1}{18}$$

E = Event of correct answer by remaining choices

$$P\left(\frac{E}{A}\right)=1$$

$$P\left(\frac{E}{B}\right)=\frac{1}{2}$$

$$P\left(\frac{E}{C}\right)=\frac{1}{3}$$

$$P\left(\frac{E}{D}\right)=\frac{1}{4}$$

Student gives the correct answer, then the probability that he knows the answer

$$P\left(\frac{A}{E}\right)=\frac{P(A)P\left(\frac{E}{A}\right)}{P(A)P\left(\frac{E}{A}\right)+P(B)P\left(\frac{E}{B}\right)+P(C)P\left(\frac{E}{C}\right)+P(D)P\left(\frac{E}{D}\right)}$$

$$=\frac{\frac{2}{3}\times 1}{\frac{2}{3}\times 1+\frac{1}{6}\times\frac{1}{2}+\frac{1}{9}\times\frac{1}{3}+\frac{1}{18}\times\frac{1}{4}}=\frac{144}{173}=\frac{a}{b}$$

Hence a + b = 144 + 173 = 317

20. (a) An urn contains 'm' white and 'n' black balls. A ball is drawn at random and is put back into the urn along with K additional balls of the same colour as that of the ball drawn. A ball is again drawn at random. What is the probability that the ball drawn now is white.

(b) An unbiased die, with faces numbered 1, 2, 3, 4, 5, 6 is thrown n times and the list of n numbers showing up is noted. What is the probability that among the numbers 1, 2, 3, 4, 5, 6, only three numbers appear in the list.

; द्धएक पात्र में 'm' सफेद और 'n' काली गेंदे है। एक गेंद यदृच्छया निकाली जाती है और उसे उसी रंग की K अतिरिक्त गेंदों के साथ पात्र में वापस रख दिया जाता है। पुनः एक गेंद यदृच्छया निकाली जाती है। निकाली गई गेंद के सफेद होने की प्रायिकता ज्ञात कीजिए।

(b) एक निष्पक्षपाती पाँसे को, जिसके फलकों पर संख्याएँ 1, 2, 3, 4, 5, 6 अंकित हैं, n बार फेंका जाता है और उस पर आने वाली n संख्याओं को नोट कर लिया जाता है। संख्याओं 1, 2, 3, 4, 5, 6 में से मात्र तीन संख्याओं के आने की प्रायिकता क्या है?

Ans. (a) $\frac{m}{m+n}$; (b) $\frac{{}^6C_3\left(3^n-3.2^n+3\right)}{6^n}$

Sol. (i) P (White Ball) = P (First drawn ball is white, then again draw a white ball after additional K white balls) + P (First drawn ball is black, then again draw a white ball after additional K black balls)

$$=\left(\frac{m}{m+n}\right)\times\left(\frac{m+k}{m+n+k}\right)+\frac{n}{m+n}\times\frac{m}{m+n+k}$$

$$=\frac{m}{m+n}$$

(ii) The total no. of outcomes = 6^n

n(E) = No. of only three number appear is the list

= ${}^6C_3$ (Total outcomes of 3 numbers – total outcomes when exactly 2 numbers appear – total outcomes when exactly 1 number appears

= ${}^6C_3$ (3^n – ${}^3C_2$ (2^n – 2) – 3)

= ${}^6C_3$ (3^n – $3\cdot2^n$ + 3)

Required probability = $\frac{{}^6C_3\left(3^n-3\cdot2^n+3\right)}{6^n}$

21. In each of a set of games it is 2 to 1 in favour of the winner of the previous game. What is the chance that the player who wins the first game shall win three at least, of the next four?

खेल के प्रत्येक सेट में, पहले वाले खेल के विजेता के पक्ष में 2 से 1 है। इस बात की कितनी संभावना है कि जो प्रथम खेल जीतता है, वही vkusoky s4 खेलों में कम से कम तीन जीतेगा?

Ans. 4/9

Sol. P(W/W) = $\frac{2}{3}$; P(L/W) = $\frac{1}{3}$; P(W/L) = $\frac{1}{3}$; P(L/L) = $\frac{2}{3}$

Favourable WWWW WWWL WWLW WLWW LWWW

Probability $\frac{2}{3}\times\frac{2}{3}\times\frac{2}{3}\times\frac{2}{3}+\frac{2}{3}\times\frac{2}{3}\times\frac{2}{3}\times\frac{1}{3}+\frac{2}{3}\times\frac{2}{3}\times\frac{1}{3}\times\frac{1}{3}+\frac{2}{3}\times\frac{1}{3}\times\frac{1}{3}\times\frac{2}{3}+\frac{1}{3}\times\frac{1}{3}\times\frac{2}{3}\times\frac{2}{3}$

$$=\frac{36}{81}=\frac{4}{9}$$

22. If mn coins have been distributed into m purses, n into each find

(1) the chance that two specified coins will be found in the same purse, and

(2) what the chance becomes when r purses have been examined and found not to contain either of the specified coins.

यदि *mn* सिक्को को *m* बटुओं में वितरित किया जाता है। प्रत्येक में n सिक्के है, तो ज्ञात करो।

(1) दो विशेष सिक्को को एक बटुए में पाये जाने की संभावना होगी

(2) इस बात की संभावना कितनी होगी यदि r बटुओं के परीक्षण के बाद किसी भी पर्स में दो विशिष्ट सिक्को में से कोई भी न मिले।

Ans. (1) $\frac{n-1}{mn-1}$, (2) $\frac{n-1}{mn-rn-1}$

Sol. Represent the two coins by A and B, then the chance that B is with A is $\frac{n-1}{mn-1}$ for whenever A is placed, there remain mn – 1 possible position for B, n – 1 of which are favourable.

Hence the chance that A and B arew not together is $\frac{n(m-1)}{mn-1}$

Now consider the m – r purses which have not been examined. If A and B are together, the chance that they occur in these purses is $\frac{m-r}{m}$

If A and B are apart the chance that occur in these purses is $\frac{(m-r)(m-r-1)}{m(m-1)}$;

For m(m – 1) is the total no. of ways in which they can occur separately in any 2 purses whatever and (m – r) (m – r – 1) is the no. of ways in which in which they can occur separately in any two of the purses we are considering

Hence the required chance, $= \frac{n-1}{mn-1}\cdot\frac{m-r}{m} \div \left\{\frac{n-1}{mn-1}\cdot\frac{m-r}{m} + \frac{n(m-1)}{mn-1}\cdot\frac{(m-r)(m-r-1)}{m(m-1)}\right\}$

$= \frac{n-1}{mn-nr-1}$

Paragraph for question nos. 23 to 25

A JEE aspirant estimates that she will be successful with an 80 percent chance if she studies 10 hours per day, with a 60 percent chance if she studies 7 hours per day and with a 40 percent chance if she studies 4 hours per day. She further believes that she will study 10 hours, 7 hours and 4 hours per day with probabilities 0.1, 0.2 and 0.7, respectively.

एक JEE की परीक्षार्थी गणना करती है कि यदि वह 10 घण्टे प्रतिदिन पढ़ती है तो उसके सफल होने की संभावना 80 प्रतिशत होगी, यदि वह 7 घण्टे पढ़ती है तो उसके सफल होने की संभावना 60 प्रतिशत होगी तथा वह 4 घण्टे प्रतिदिन पढ़ती है तो उसके सफल होने की l ikkouk40 प्रतिशत होगी वह जानती है कि उसके 10 घण्टे, 7 घण्टे एवं 4 घण्टे प्रतिदिन पढ़ने की संभावनाएं क्रमशः 0.1, 0.2 तथा 0.7 है।

23. The chance she will be successful, is :

उसके सफल होने की संभावना है :

(A) 0.28 (B) 0.38 (C) 0.48 (D) 0.58

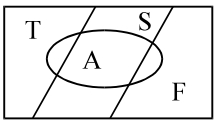
Ans. C

Sol. A : She get a success

T : She studies 10 hrs : P(T) = 0.1

S : She studies 7 hrs : P(S) = 0.2

F : She studies 4 hrs : P(F) = 0.7



P(A/T) = 0.8 ; P(A/S) = 0.6 ; P(A/F) = 0.4

$P(A) = P(A \cap T) + P(A \cap S) + P(A \cap F)$

$= P(T) \cdot P(A/T) + P(S) \cdot P(A/S) + P(F) \cdot P(A/F)$

$= (0.1)(0.8) + (0.2)(0.6) + (0.7)(0.4)$

$= 0.08 + 0.12 + 0.28 = 0.48$

24. Given that she is successful, the chance she studied for 4 hours, is :

यदि वह सफल होती है, तो उसके 4 घण्टे पढ़ने की संभावना होगी :

(A) $\frac{6}{12}$ (B) $\frac{7}{12}$ (C) $\frac{8}{12}$ (D) $\frac{9}{12}$

Ans. B

Sol. $P(F/A) = \frac{P(F \cap A)}{P(A)} = \frac{(0.7)(0.4)}{0.48} = \frac{0.28}{0.48} = \frac{7}{12}$

25. Given that she does not achieve success, the chance she studied for 4 hour, is :

यदि वह असफल होती है, तो इसकी संभावना कि उसने 4 घण्टे पढ़ाई की होगी :

(A) $\frac{18}{26}$ (B) $\frac{19}{26}$ (C) $\frac{20}{26}$ (D) $\frac{21}{26}$

Ans. D

Sol. $P(F/\overline{A}) = \frac{P(F \cap \overline{A})}{P(\overline{A})} = \frac{P(F) - P(F \cap A)}{0.52} = \frac{(0.7) - 0.28}{0.52} = \frac{0.42}{0.52} = \frac{21}{26}$

Paragraph for Question Nos. 26 to 28

A box contains b red balls, '2b' white balls and '3b' blue balls where *b* is a positive integer. 3 balls are selected at random from the box.

एक बक्से में b लाल गेंदे, '2b' सफेद तथा '3b' नीली गेंदे है। जहाँ *b* धनात्मक पूर्णांक है। तीन गेंदो को यादृच्छया निकाला जाता है।

26. If balls are drawn without replacement and 'A' denotes the event that "No two of the selected balls have the same colour" then :

(A) there is no value of b for which P(A) = 0.3

(B) There is exactly one value of b for which P(A) = 0.3 and this value is less than 2.

(C) There is exactly one value of b for which P(A) = 0.3 and this value is greater than 2.

(D) There is more than one value of b for which P(A) = 0.3

यदि गेदें यादृच्छया निकाली जाती है (पुर्नस्थापना के बिना) एवं 'A' घटना ''कोई भी दो निकाली गयी गेदें समान रंग की नहीं है'' को i nf' kZ d j r hg\$ r ks

(A) च; । ढ> 0ष्ठ के लिये b का कोई मान नही है

(B) च; । ढ> 0ष्ठ के लिये b का ठीक एक मान होगा जो कि 2 से छोटा होगा

(C) च; । ढ> 0ष्ठ के लिये b का ठीक एक मान होगा जो कि 2 से बड़ा होगा

(D) च; । ढ> 0ष्ठ के लिये b के एक से ज्यादा मान होगे

Ans. B

Sol. Bag $\begin{cases} b \text{ red balls} \\ 2b \text{ white balls} \\ 3b \text{ blue balls} \end{cases}$

$$P(A) = P(\text{all different colour}) = \frac{{}^{b}C_1 \cdot {}^{2b}C_1 \cdot {}^{3b}C_1}{{}^{6b}C_3} = \frac{6b^3 \cdot 6}{6b(6b-1)(6b-2)} = P(A)$$

$$P(A) = \frac{6b^2}{(6b-1)(6b-2)} = \frac{3b^2}{(6b-1)(3b-1)} = \frac{3}{10}$$

$10b^2 = 18b^2 - 9b + 1$

$8b^2 - 9b + 1 = 0 \quad \Rightarrow \quad (b-1)(8b-1) = 0$

$b = 1$ Ans.

27. If balls are drawn without replacement and 'B' denotes the event that "No two of the 3 drawn balls are blue" then:

(A) $P(B) = \frac{1}{3}$ if b=1 (B) $P(B) = \frac{2}{3}$ if b=2 (C) $P(B) = \frac{1}{4}$ if b=4 (D) $P(B) = \frac{1}{2}$ for all value of b.

यदि गेंदो को बिना पुर्नस्थापना के निकाला जाता है तथा B यदि निकाली गयी 3 गेंदो में से, कोई भी दो के नीली ना होने की प्रायिकता gks h:

(A) $P(B) = \frac{1}{3}$ यदि b=1 (B) $P(B) = \frac{2}{3}$ यदि b=2 (C) $P(B) = \frac{1}{4}$ यदि b=4 (D) च; ठ ढ> $\frac{1}{2}$ b के मानों के लिये

Ans. D

Sol. P(B) = P(no two of them are blue)

= 1 – P[(B B B) or B B and one R or W) i.e. BBB or BBR or BBW]

$$= 1 - \left[\frac{^{3b}C_3}{^{6b}C_3} + \frac{^{3b}C_2 \cdot {}^{3b}C_1}{^{6b}C_3} \right] = 1 - \left[\frac{3b(3b-1)(3b-2)}{6b(6b-1)(6b-2)} + \frac{3b(3b-1)3b \cdot 6}{2 \cdot 6b(6b-1)(6b-2)} \right]$$

$$= 1 - \frac{1}{4(6b-1)} \left[(3b-2) + 9b \right]$$

$$= 1 - \frac{2(6b-1)}{4(6b-1)} = \frac{1}{2}$$

hence P(B) is independent b

28. If P(A) = 0.3, then the value of P(A/B) equals :
यदि P(A) = 0.3 तो P(A/B) का मान होगा :
(A) 3/5 (B) 3/10 (C) 1/2 (D) 2/3

Ans. A

Sol. $P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)}{P(B)}$ (think !)

$= \frac{3}{10} \cdot \frac{2}{1} = \frac{3}{5}$ Ans.

Paragraph for Question Nos. 29 to 31

Urn-I contains 5 Red balls and 1 Blue ball,
Urn-II contains 2 Red balls and 4 Blue balls.
A fair die is tossed. If it results in an even number, balls are repeatedly withdrawn one at a time with replacement from urn-I. If it is an odd number, balls are repeatedly withdrawn one at a time with replacement from urn-II. Given that the first two draws both have resulted in a blue ball.

थैला–I में 5 लाल तथा 1 नीली गेंदें है।
थैला–II में 2 लाल तथा 4 नीली गेंदे है।
एक पांसे को उछाला जाता है। यदि सम संख्या आती है तो गेंदो को थैला-I से पुर्नस्थापना के साथ बार–बार निकाला जाता है। यदि विषम संख्या आती है तो गेंदो को थैला–II से बार बार निकाला जाता है तथा पुर्नस्थापित कर दिया जाता है। प्रथम दो निकासी से नीली गेंद निकलती है।

29. Conditional probability that the first two draws have resulted in blue balls given urn-II is used is :
(A) 1/2 (B) 4/9 (C) 1/3 (D) None

थैला–II के प्रयुक्त होने पर प्रथम दो निकासी से नीली गेंद आने की प्रायिकता होगी :
(A) 1/2 (B) 4/9 (C) 1/3 (D) कोई नहीं

Ans. B

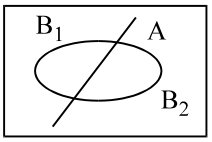
Sol. Urn-I $\begin{cases} 5R \\ 1B \end{cases}$

Urn-II $\begin{cases} 2R \\ 4B \end{cases}$

A: first two draws resulted in a blue ball.

B_1 : urn-I is used $\quad P(B_1) = \frac{1}{2}$

B_2 : urn-II is used $\quad P(B_2) = \frac{1}{2}$



$$P(A/B_1) = \frac{1}{6}\cdot\frac{1}{6} = \frac{1}{36}$$

$$P(A/B_2) = \frac{4}{6}\cdot\frac{4}{6} = \frac{16}{36} = \frac{4}{9}$$ Ans.

30. If the probability that the urn-I is being used is *p*, and *q* is the corresponding figure for urn-II then :

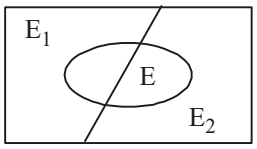
यदि थैला–I के प्रयुक्त होने की प्रायिकता *p* तथा थैला–II के प्रयुक्त होने की प्रायिकता q हो तो :

(A) q = 16p (B) q = 4p (C) q = 2p (D) q = 3p

Ans. A

Sol.

$$\left.\begin{aligned} \underbrace{P(B_1/A)}_{E_1} &= \frac{\frac{1}{2}\cdot\frac{1}{36}}{\frac{1}{2}\cdot\frac{1}{36}+\frac{1}{2}\cdot\frac{16}{36}} = \frac{1}{17} \\ \underbrace{P(B_2/A)}_{E_2} &= \frac{\frac{1}{2}\cdot\frac{16}{36}}{\frac{1}{2}\cdot\frac{16}{36}+\frac{1}{2}\cdot\frac{1}{36}} = \frac{16}{17} \end{aligned}\right] \Rightarrow \text{Ans.}$$



31. The probability of getting a red ball in the third draw, is :

तीसरी निकासी पर लाल गेंद आने की प्रायिकता होगी :

(A) 1/3 (B) 1/2 (C) 37/102 (D) 41/102

Ans. C

Sol. E : third ball drawn is red

$P(E) = P(E \cap E_1) + P(E \cap E_2)$

$$= \frac{1}{17}\cdot\frac{5}{6}+\frac{16}{17}\cdot\frac{2}{6} = \frac{5}{102}+\frac{32}{102} = \frac{37}{102}$$ Ans.

Paragraph for Question Nos. 32 to 33

In world cup 2007, 16 teams participating including team A, B and C. Their probability of wining the world cup are 1/4, 1/6, 1/2 respectively and remaining all the teams has equal probability to win. Now two persons X and Y has probability of speaking the truth 1/2 and 1/3 respectively. After the final is over, some one who has not seen the match asked both X and Y (who has seen the match), that "which team has won the world cup?" (given that match ends with proper result)

विश्व कप 2007 में, 16 टीमों ने भाग लिया जिनमें टीम A, B तथा C शामिल थी। इनकी जीतने की प्रायिकतायें क्रमशः 1/4, 1/6, 1/2 है। तथा शेष अन्य टीमों की जीतने की प्रायिकतायें समान है। अब दो व्यक्ति X तथा Y जिनके सत्य होने की प्रायिकतायें क्रमशः 1/2 तथा 1/3 है। जब अन्तिम मैच समाप्त हो जाता है तब एक व्यक्ति जिसने मैच नहीं देखा था, दोनों व्यक्तियों X तथा Y (जिन्होनें मैच देखा Hkk) को पूछता है कि "विश्व कप कौनसी टीम ने जीता है ?"

(दिया गया है अन्तिम मैच एक टीम जीतती है)

32. The probability that both X and Y will say that team A is the winner is :

(A) 77/1800 (B) 7/24 (C) 1/24 (D) None of these

वह प्रायिकता जब दोनों X तथा Y ने कहा कि विश्व कप A टीम ने जीता है :

(A) 77/1800 (B) 7/24 (C) 1/24 (D) इनमें से कोई नहीं

Ans. A

Sol. E_1 : A will win

E_2 : A doesn't win

E : Both X & Y say team A is winner

$P(E) = P(E_1 \cap A) + P(E_2 \cap A)$

$$= P(E_1)\cdot P\left(\frac{A}{E_1}\right) + P(E_2)\cdot P\left(\frac{A}{E_2}\right)$$

$$= \frac{1}{4}\times\frac{1}{2}\times\frac{1}{3}+\frac{3}{4}\times\frac{1}{2}\times\frac{2}{3}\times\frac{1}{15}\times\frac{1}{15}$$

$$= \frac{77}{1800}$$

33. If both X and Y has said that team A has won the world cup, then the probability that team A has actually won the world cup is :

(A) 75/77 (B) 1/7 (C) 1 (D) None of these

यदि दोनों X तथा Y ने कहा कि टीम A ने विश्व कप जीता है, तब टीम A द्वारा वास्तव में विश्व कप जीतने की प्रायिकता है :

(A) 75/77 (B) 1/7 (C) 1 (D) इनमें से कोई नहीं

Ans. A

Sol. $P\left(\frac{E_1}{E}\right)=\frac{P(E_1 \cap E)}{P(E)}$

$$=\frac{\frac{1}{4}\times\frac{1}{2}\times\frac{1}{3}}{\frac{77}{1800}}=\frac{75}{77}$$

MATCH THE COLUMN

34. Match the following columns :

Column-I	**Column-II**
(A) Two different numbers are taken from the set {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10}. The probability that their sum and positive difference, are both multiple of 4, is x/55 then *x* equals	(P) 4
	(Q) 6
(B) There are two red, two blue, two white and certain number (greater than 0) of green socks in a drawer. If two socks are taken at random from the drawer without replacement, the probability that they are of the same colour is 1/5 then the number of green socks are	(R) 8
(C) A drawer contains a mixture of red socks and blue socks, at most 17 in all. It so happens that when two socks are selected randomly withoutreplacement, there is a probability of exactly 1/2 that both are red or both are blue. The largest possible number of red socks in the drawer that is consistent with this data, is	(S) 10

निम्न कॉलम को सुमेलित कीजिए :

Column-I	**Column-II**
(A) समुच्चय {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10} से दो भिन्न संख्याओं को निकाला जाता है। उनके योग एवं धनात्मक अन्तर दोनों के 4 का गुणक होने की प्रायिकता x/55 हो तो *x* होगा	(P) 4
	(Q) 6
(B) किसी मेज की दराज में 2 लाल, 2 नीली, 2 सफेद तथा कुछ हरे (> 0) मोजे है। यदि दो मोजो की यादृच्छया निकाला जाता है, बिना पुर्नस्थापना के दो मोजो को निकाला जाता है तो उनके समान रंग होने की प्रायिकता 1/5 है, तो हरे रंग के मोजो की संख्या होगी	(R) 8

(C) किसी दराज में लाल एवं नीले मोजे है जिनकी अधिकतम संख्या कुल 17 है। यदि दो मोजो को यादृच्छया निकाला जाता है बिना पुर्नस्थापना के, तो दोनों के लाल या दोनों के नीले रंग के होने की 1/2 है अतः दराज में रखे लाल मोजो की अधिकतम संभावित संख्या होगी (S) 10

Ans. A–Q, B–P, C–S

Sol. (A) Let the two numbers are 'a' and 'b'

$$\left.\begin{matrix} a+b=4p \\ a-b=4q \end{matrix}\right] \quad p,q \in I$$

$2a = 4(p+q) \Rightarrow a = 2I_1$

$2b = 4(p-q) \Rightarrow b = 2I_2$

Hence both a and b must be even. Also note that if $(a-b)$ is a multiple of 4 then $(a+b)$ will automatically be a multiple of 4.

Hence $n(S) = {}^{11}C_2$

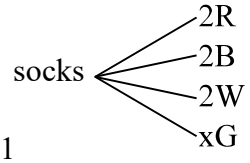
$n(A) = (0, 4), (0, 8), (2, 6), (2, 10), (4, 8), (6, 10) = 6$

$$\therefore \quad P(A) = \frac{6}{{}^{11}C_2}$$

(B) Let the number of green socks are $x > 0$

E : two socks drawn are of the same colour

P(E) = P(R R or B B or W W or G G)



$$= \frac{3}{{}^{6+x}C_2} + \frac{{}^{x}C_2}{{}^{6+x}C_2} = \frac{6}{(x+6)(x+5)} + \frac{x(x-1)}{(x+6)(x+5)} = \frac{1}{5}$$

$5(x^2 - x + 6) = x^2 + 11x + 30$

$4x^2 - 16x = 0 \Rightarrow x = 4$ Ans.

(C) Let there be x red socks and y blue socks. Then $\frac{{}^{x}C_2 + {}^{y}C_2}{{}^{x+y}C_2} = \frac{1}{2}$

let $x > y$

or $$\frac{x(x-1)+y(y-1)}{(x+y)(x+y-1)} = \frac{1}{2}$$

Multiplying both sides by $2(x+y)(x+y-1)$ and expanding,

we find that $2x^2 - 2x + 2y^2 - 2y = x^2 + 2xy + y^2 - x - y$.

Rearranging, we have $x^2 - 2xy + y^2 = x + y \Rightarrow (x-y)^2 = x + y \Rightarrow |x-y| = x + y$

Since $x + y \le 17$, $x - y \le \sqrt{17}$. as $x - y$ must be an integer $\Rightarrow x - y = 4$

$\therefore \quad x + y = 16$

Adding both together and dividing by two yields $x \le 10$ Ans.

35. Match the following columns :

Column-I	**Column-II**
(A) The probability of a bomb hitting a bridge is 1/2. Two direct hits are needed to destroy it. The least number of bombs required so that the probability of the bridge being destroyed is greater than 0.9, is	(P) 4
(B) A bag contains 2 red, 3 white and 5 black balls, a ball is drawn its colour is noted and replaced. Minimum number of times, a ball must be drawn so that the probability of getting a red ball for the first time is at least even, is	(Q) 5
(C) A hunter knows that a deer is hidden in one of the two near by bushes, the probability of its being hidden in bush-I being 4/5. The hunter having a rifle containing 10 bullets decides to fire them all at bush-I or II. It is known that each shot may hit one of the two bushes, independently of the other with probability 1/2. Number of bullets must he fire on bush-I to hit the animal with maximum probability is (Assume that the bullet hitting the bush also hits the animal).	(R) 6
	(S) 7

निम्न कॉलम को सुमेलित कीजिए :

Column-I	**Column-II**
(A) किसी बम के पुल से टकराने की प्रायिकता 1/2 है। इसे उड़ाने के लिये दो सीधे टक्करो की जरूरत पड़ती है। कम से कम कितने बमो की जरूरत पड़ेगी यदि पुल के उड़ने की प्रायिकता 0.9 से अधिक हो	(P) 4
(B) किसी थैले में 2 लाल, 3 सफेद तथा 5 काली गेंदे है एक गेंदे निकाली जाती है,इसका रंग नोट किया जाता है और वापस रख दिया जाता है। कम से कम कितनी बार गेंद निकाली जाय कि प्रथम बार लाल गेंद आने की प्रायिकता ½ या ज्यादा हो	(Q) 5
(C) एक शिकारी जानता है कि एक हिरण पास की दो में से किसी एक झाड़ी में छुपा हुआ है इसके झाड़ी I में छुपने की प्रायिकता 4/5 है। शिकारी के पास राइफल जिसमें 10 गोलियां है जिन्हें वह झाड़ी I या II में दागना चाहता है। प्रत्येक शाट स्वतंत्रता पूर्वक किसी एक झाड़ी से 1/2 की प्रायिकता से टकरा जाता है। यदि हिरण को मारने के लिये प्रायिकता अधिकतम हो, तो झाड़ी I में दागी गई गोलियों d hl fn; k(माना जा सकता है कि झाड़ी में दागी गई गोली हिरण से भी टकराती है)	(R) 6
	(S) 7

Ans. A – S, B –P, C – R

Sol. (A) P(S) = 1/2 ; P(F) = 1/2

Let 'n' be the least number of bombs to be dropped

E : bridge is destroyed $\Rightarrow$ P(E) = 1 – P(0 or 1 successes)

$$= 1 - \left(\left(\frac{1}{2}\right)^n + {}^nC_1 \cdot \frac{1}{2} \cdot \left(\frac{1}{2}\right)^{n-1} \right) = 1 - \left(\frac{1}{2^n} + \frac{n}{2^n} \right) \geq 0.9$$

or $\frac{1}{10} \geq \frac{n+1}{2^n}$ or $\frac{2^n}{10(n+1)} \geq 1$

The value of n consistent with n = 7 or draw graph between $y = 2^x$ and $y = 10(x+1)$.

(B) Bag $\begin{cases} 2\ R \\ 3\ B \\ 5\ B \end{cases}$; $P(S) = \frac{1}{5}$; $P(F) = \frac{4}{5}$; E : getting a red ball

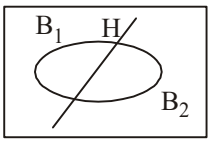
$P(E) = P(S \text{ or } F\ S \text{ or } F\ F\ S \text{ or } .....] \geq \frac{1}{2}$; hence $\frac{P(S)[1-(P(F))^n]}{1-P(F)} \geq \frac{1}{2}$

$P(F)^n \leq \frac{1}{2}$; $\left(\frac{4}{5}\right)^n \leq \frac{1}{2}$

The value of n consistent with this is 4 $\Rightarrow$ (P)

(C) B_1 : animal hides in Bush I

B_2 : animal hides in Bush II



$P(B_1) = \frac{4}{5}$; $P(B_2) = \frac{1}{5}$; $P(H) = \frac{1}{2}$

Let ξ βυλλετσ → Bush I;

Let 10 − ξ βυλλετσ → Bush II;

$$P(H) = \frac{4}{5}\underbrace{\left[1-\left(\frac{1}{2}\right)^x\right]}_{P(\text{at least one hit})} + \frac{1}{5}\underbrace{\left[1-\left(\frac{1}{2}\right)^{10-x}\right]}_{P(\text{at least one hit})}$$

$$\frac{dP}{dx} = -\frac{4}{5}\left(\frac{1}{2}\right)^x \ln\frac{1}{2} + \frac{1}{5}\left(\frac{1}{2}\right)^{10-x} \ln\left(\frac{1}{2}\right) = 0 \quad \Rightarrow \quad x = 6$$

36. A bag contains some white and some black balls. All combinations being equally likely. The total number of balls in the bag is 12. Four ball are drawn at random one by one from the bag without replacement :

Column-I	**Column-II**
A. Probability that all the four balls are black is equal to	P. $\frac{14}{33}$
B. If the bag contains 10 black and 2 white balls then the probability that all four balls are black is equal to	Q. $\frac{1}{5}$
C. If all four balls drawn are found to be black then the Probability that the bag contains 10 black balls is equal to	R. $\frac{70}{429}$

D. Probability that two balls are black and two are white given bag contains equal number of white and black balls — S. $\frac{5}{11}$

एक थैले में कुछ सफेद एवं कुछ काली गेंदें रखी है। सभी संचय (combinations) समसम्भावी हैं। थैले में गेंदों की कुल संख्या 12 हो। थैले से 4 गेंदें यादृच्छिक रूप से बिना प्रतिस्थापित किये निकाली जाती है :

	Column-I	**Column-II**
A.	चारों गेंद काली होने की प्रायिकता है	P. $\frac{14}{33}$
B.	यदि थैले में 10 काली और 2 सफेद गेंद है, तो सभी चार गेंदों के काली होने की प्रायिकता है।	Q. $\frac{1}{5}$
C.	यदि सभी चार गेंदे काली पायी जाती है, तो थैले में 10 काली गेंदे होने की प्रायिकता है।	R. $\frac{70}{429}$
D.	यदि दिए गए थैले में सफेद और काली गेंदों की संख्या बराबर हो, तो दो काली गेंदें और दो सफेद गेंदें होने की प्रायिकता है।	S. $\frac{5}{11}$

Ans. A – Q, B –P, C – R, D – S

Sol. (B) If the bag contains 10 black and 2 white balls, the probability that all four balls are black is equal to

$$\frac{^{10}C_4}{^{12}C_4}=\frac{8\times 7}{12\times 11}=\frac{14}{33}$$

(A) Let E_1 : an event that there are i black balls, (12–i) white balls, where i = 0, 1, 2, 3, ... 12 i.e., 13 outcomes with each combination equally likely

And A : an even that all the four balls drawn are black.

Then, $P(E_i)=\frac{1}{13}$ and $P(A/E_i)=0$ when i = 0, 1, 2, 3 as the number of black balls drawn are 4 and $P\left(\frac{A}{E_i}\right)=\frac{^{i}C_4}{^{9}C_4}$ when i = 4, 5, 6, ... 12 = 9 outcomes.

Hence $P(A)=\sum_{i=4}^{12}\left(P(E_i)\times P\left(\frac{A}{E_i}\right)\right)$

$$\Rightarrow P(A)=\frac{1}{13}\left(\frac{^{4}C_4}{^{12}C_4}+\frac{^{5}C_4}{^{12}C_4}+\frac{^{6}C_4}{^{12}C_4}+\frac{^{7}C_4}{^{12}C_4}+\frac{^{8}C_4}{^{12}C_4}+\frac{^{9}C_4}{^{12}C_4}+\frac{^{10}C_4}{^{12}C_4}+\frac{^{11}C_4}{^{12}C_4}+\frac{^{12}C_4}{^{12}C_4}\right)$$

$^{4}C_4$ can be written as $^{5}C_5$ and we know $^{n}C_r + {}^{n}C_{r-1} = {}^{n+1}C_r$

Hence ${}^4C_4 + {}^5C_4 + {}^6C_4 + {}^7C_4 + \ldots\ldots\ldots + {}^{12}C_4$

$= {}^5C_5 + {}^5C_4 + {}^6C_4 + {}^7C_4 + \ldots\ldots\ldots + {}^{12}C_4$

$= {}^6C_5 + {}^6C_4 + {}^7C_4 + \ldots\ldots\ldots + {}^{12}C_4$

$= {}^7C_5 + {}^7C_4 + \ldots\ldots\ldots + {}^{12}C_4$

$= {}^{12}C_5 + {}^{12}C_4 = {}^{13}C_5$

Therefore $P(A) = \frac{1}{13} \times \frac{1}{{}^{12}C_4} \times {}^{13}C_5 = \frac{1}{13} - \frac{13}{5} = \frac{1}{5}$

is the probability that four balls are black.

(C) Now here we need to find Probability of 10 black balls when event

A has already occured, i.e., $P\left(\frac{E_{10}}{A}\right) = \frac{P(E_{10} \cap A)}{P(A)}$

$$\Rightarrow P\left(\frac{E_{10}}{A}\right) = \frac{P(E_{10}) \times P\left(\frac{A}{E_{10}}\right)}{P(A)} = \frac{\frac{1}{13} \times \frac{14}{33}}{\frac{1}{5}} = \frac{70}{429}$$

$$= \frac{14 \times 10}{13 \times 11 \times 6} = \frac{70}{429}$$

is the required probability that the bag contains 10 black balls.

(D) $\frac{{}^6C_2 \cdot {}^6C_2}{{}^{12}C_4} = \frac{5}{11}$